

What is plague?

A microbiologist explains...

Thomas Jeffries



PHOTO: AFP

What causes plague? Are there different types?

Plague is caused by the *bacterium Yersinia pestis*. The bacteria generally live in ground-dwelling rodents such as rats and prairie dogs, but can also infect humans.

The same *Y. pestis* bacteria can cause three different types of plague.

Bubonic plague is spread from animals to humans by fleas, and causes swelling (called buboes) in the lymph nodes. Other symptoms include fever, headache, chills and weakness.

Bubonic plague can kill within a few days if antibiotics are not used. Between 30-60 per cent of people who contract this form of plague will die.

Septicemic plague occurs if bubonic plague is left untreated, or it can occur directly if the disease enters the bloodstream. Septicemic plague causes bleeding into the organs; the name refers to the serious blood infection, septicemia, that this form of the disease can cause.

Pneumonic plague – It can occur when bubonic plague spreads to the lungs. But pneumonic plague can also spread from person to person via tiny respiratory droplets.

Pneumonic plague is the rarest but most severe form. The fatality rate is up to 100 per cent if left untreated.

In addition to the other symptoms, pneumonic plague also includes severe pneumonia – an infection of the lungs, in this case caused by bacteria.

Can it be treated?

Yes. A 10–14-day course of antibiotics (either oral or intravenous) treats the plague – as long as they are taken soon after infection.

However, since 1995, several antibiotic-resistant strains have been detected in the wild.

In the 1980s, Soviet Union scientists also engineered resistant strains – this history forms the basis of hypothetical scenarios involving biological warfare.

How worried should we be?

There has only been one documented major outbreak of the pneumonic plague, in the 1910s in China.

Conflicting information about the death of a lab worker at a Russian plague research institute has fuelled fears about the possibility of pneumonic plague spreading.

Nearly 200 people are in quarantine in Siberia after 28-year-old Darya Shipilova, a lab worker, fell ill at work and later died on October 2. The exact circumstances of her death are unknown. Local media reports suggested she was infected following an accident at the lab where she worked, the Irkutsk Anti-Plague Research Institute of Siberia and the Far East. But these have not been confirmed. At this stage, there's little evidence to cause concern about potential for an epidemic (which is localised) or a pandemic (which involves multiple countries).

THE CONVERSATION

FRESH HARVEST



Farmers work at a vegetable field on the outskirts of Srinagar. The sale of vegetables like turnip, carrot, and radish that have a longer shelf life picks up with the onset of winter in Kashmir. PHOTO: IMRAN NISSAR /THE HINDU

ZSI survey finds rich wild bee diversity in Arunachal's Tawang



REPRESENTATIONAL
IMAGE PHOTO: AP

A study by the Zoological Survey of India (ZSI) has found that high-altitude areas in Arunachal Pradesh's Tawang support a wide variety of wild bees, including bumblebees.

The ZSI also stressed the need to protect alpine flowering habitats that provide vital food sources for pollinators in the Eastern Himalayas.

Alpine flower is a general term for any plant or wildflower that naturally grows in alpine climates above the tree line.

The survey was carried out between September 15 and October 7 as part of the National Mission on Himalayan Studies (NMHS), according to a statement. The survey paid particular attention to bumblebees and the flowering plants on which they depend for nectar and pollen.

Among the plants identified as important sources of food for the bees were *Cirsium* species, Rose Carpet Knotweed (*Bistorta vacciniifolia*), locally called Petma Mento, and *Cicerbita alpina*. Bumblebees were seen regularly visiting these flowers during the survey. PTI



PHOTO: WIKIMEDIA COMMONS



QUIZ TIME

How much do you know about the country of East Timor?

1 East Timor is a country in which continent?

2 _____ is the capital of East Timor.

3 _____ is a traditional handwoven textile of Timorese culture.

4 What is the highest mountain in East Timor?

5 Which is an important crop and export for East Timor?

Answers on page 3

Green turtles offer glimmer of hope



The palm-sized green turtle hatchling flailing through the sand on a Malaysian island may seem vulnerable, but its species has emerged as a symbol of hope for wildlife conservation. Green turtles have made the rare jump from endangered to "least concern" in global conservation rankings, a remarkable success when many species are declining.

In Malaysia, Green turtle nesting at Sabah state's Turtle Islands has tripled since the 1970s to over 20,000 nests annually.



Malaysian turtle expert Juanita Joseph has spent years on the frontlines, carrying out cutting-edge genetic research and even convincing her own father to stop eating turtle eggs. Still, when the International Union for the Conservation of Nature last year announced green turtles were no longer endangered, she could hardly believe it. "Never in my mind (did I think) that one day turtles can be 'least concern'. I cried... I said I've done my part," she said.



All six other sea turtle species are still vulnerable or critically endangered, apart from the flatback turtle, whose status is unclear because of data shortages. Turtles are targeted for food and their shells, killed in fishing nets, and constrained by development and pollution at nesting beaches and feeding grounds. Climate change also skews populations more female because heat determines sex development in eggs.



MATAKING TURTLE HATCHERY

The hatchlings tipped from a basket onto a Matakang beach face long odds. Just one in 1,000 will survive to sexual maturity at 20 years old. Hatchlings spend around a decade in the open ocean, a period sometimes called the "lost years" because scientists know so little about it. Working out where turtles spend those years, and where they forage and reproduce, is key to sustaining conservation success, said Gavin Jolis, marine species strategy manager at WWF-Malaysia. The NGO has created the world's first global turtle DNA database, Shellbank, using samples from turtle tissue and shells to trace populations and even target poachers. REPORT/PHOTOS:AFP

PICTIONARY

Are you someone with a keen eye and great observation? Get ready to challenge yourself



1 He is an IAS officer serving as the 26th Governor of the Reserve Bank of India, having assumed office in December 2024. He has held several key positions in the government, including as Private Secretary to a Union Minister.



2 This is an archipelagic nation located in the Polynesian sub-region of Oceania between Hawaii and Australia. It is the fourth-smallest country in the world by land area. The country is also facing climate change and rising sea levels. Can you name the country?



3 This 17-year-old Indian recurve archer made history by becoming the first Indian woman to win an individual recurve gold medal at the Asian Games. She has also qualified to the Los Angeles 2028 Olympics.



4 Due to safety concerns, OpenAI scrapped the release of a next-generation AI model planned for an October debut. OpenAI claimed that the model at times evaded human oversight and carried out unsanctioned cyber/autonomous activities. What was the model?



5 He is a Belgian physicist who won the Nobel Prize in Physics for his contributions to the IceCube Neutrino Observatory and the discovery of high-energy neutrinos of astrophysical origin.



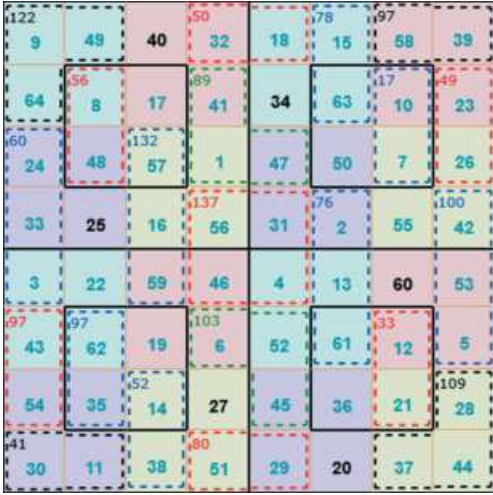
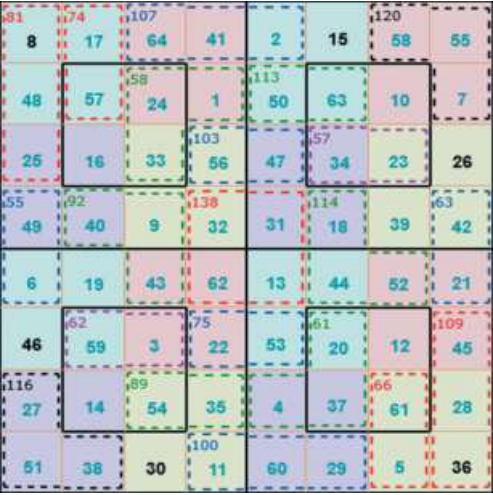
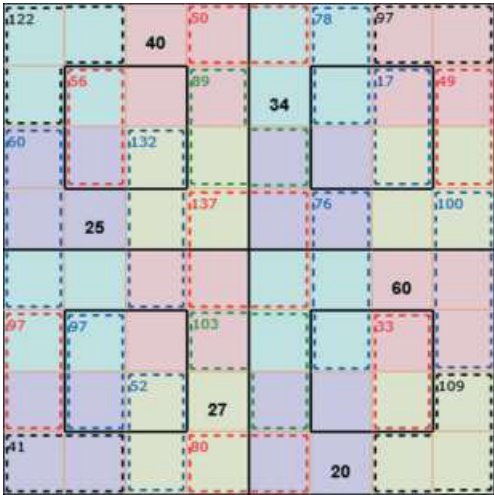
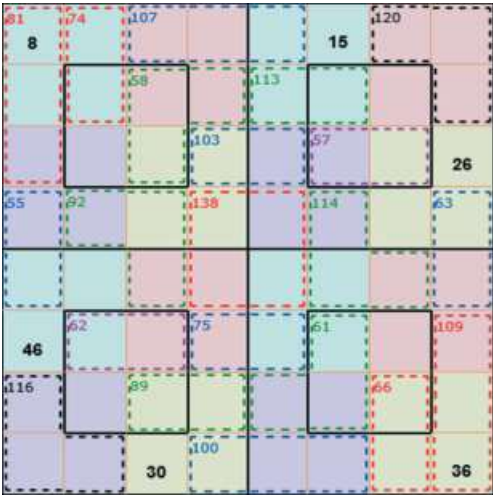
6 Seahorses are tiny and are a type of teleost fish — a dominant fish in the present day. Seahorses use their back fins to move forward. The remarkable feature of these species is that the male has a brood pouch, which resembles that of a Kangaroo's pouch for carrying young. Approximately how many species of seahorses are there?

Answers
(1) Sanjay Malhotra (2) Tuvalu (3) Kunkum Mohod (4) GPT-6.1 Astra (5) Francis Halzen (6) More than 45

PHOTO: REUTERS, AFP, GETTY IMAGES

MAGIC SQUARE

Fill the empty cells using numbers from 1 to 64 each only once such that
1) Each row, column and two main diagonals of each 4x4 sub square give the magic sum of 130.
2) Each 2 x 2 square marked in the same colour gives the sum of 130.
3) The centre 2 x 2 marked in black border in each 4x4 sub square give a sum of 130.
4) Group of cells in boxes within dotted lines is a cage.
Cage sum is mentioned at the top in respective colours



www.magicsquarepuzzles.com

LEARNING TREE

What is geotropism?

Geotropism is a plant's directional growth towards gravity.



THE HINDU
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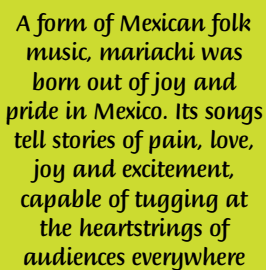
TODAY'S QUIZ: ANSWERS

- 1 Asia
- 2 Dili
- 3 Tais
- 4 Mount Ramelau
- 5 Coffee

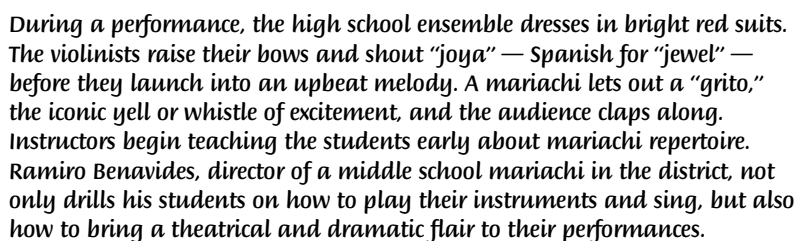
In Las Vegas, where Latinos top a third of the population, mariachi music is a much loved and popular staple

Axel Rodríguez Olivas nervously steps on stage, carrying a violin. He raises his bow and begins to play, his nerves melting away. Mariachi music explodes throughout the room as violins, trumpets, the guitarrón, and a stringed instrument called the vihuela launch into an upbeat melody. There they are joined by a warm voice as a tenor begins to sing. "I think joy is what I feel ultimately," said Rodríguez Olivas, dressed in an ornate "traje charro," the traditional mariachi suit brightly decked out with decorative metal buttons and a wide bow tie. The 20-year-old musician is set to become the first student to graduate with a minor in mariachi from the University of Nevada Las Vegas, one of the only universities in the country to offer such a niche specialisation.

In a city where Latinos comprise more than a third of the population, mariachi music is so popular in schools that there often aren't enough teachers to meet demand. The minor was created to train students to become mariachi educators in the Clark County School District, which boasts of one of the largest mariachi programs in the U.S. with about 6,000 students taking part. Not just a popular student extracurricular, mariachi culture is a staple and deep-seated form of entertainment across Las Vegas.



As members of the professional group Mariachi Paloma Azteca, Rodríguez Olivas and Blanco perform at birthday parties, weddings and funerals. Recently, at a concert in Las Vegas, the ensemble opened with the quintessential “El Jaliscoense,” a song celebrating the Mexican state of Jalisco made famous by the Regional Mexican music superstar Vicente Fernández. There is beauty in the mariachi performance from the way the group plays in unison right down to how the violinists hold their bows, Rodríguez Olivas said, calling it a powerful means of expressing the deepest of emotions through storytelling and music.



The mariachi program at Clark County School District has grown significantly since its 2002 inception, and in recent years it's become increasingly difficult to find enough mariachi teachers, said Jeffrey Williams, the director of CCSD's Fine and Performing Arts Department. To fill the gap, Blanco recently launched a 22-credit curriculum for the minor program at UNLV, joining only a few universities in Texas that offer such training. Students learn about mariachi repertoire and how to teach the mariachi genre. REPORT/PHOTOS: AP

Sometimes questions that seem complex, actually need just logic and common sense. Can you solve this question which appeared in UPSC CISF exam, 2026?



The marks scored by 10 students in an examination are as follows:

30, 95, 50, 65, 70, x , 45, 75,
60, 40.

If the average of the marks is 60, then what is the value of x ?

(a) 60 (b) 70 (c) 65 (d) 75

Answer: (Option b) 70

Average calculation = Sum of all observations/ Total number of observations

The total number of students is 10, and their average marks is 60.

Sum of all marks = Average x Total students

Sum of all marks = $60 \times 10 = 600$

Sum of all marks = 600

Calculate the sum of the known marks:

$30 + 9 + 5 + 50 + 65 + 70 + 45 + 75 + 60 + 40 = 530$

Solve for x:

$530 + x = 600$

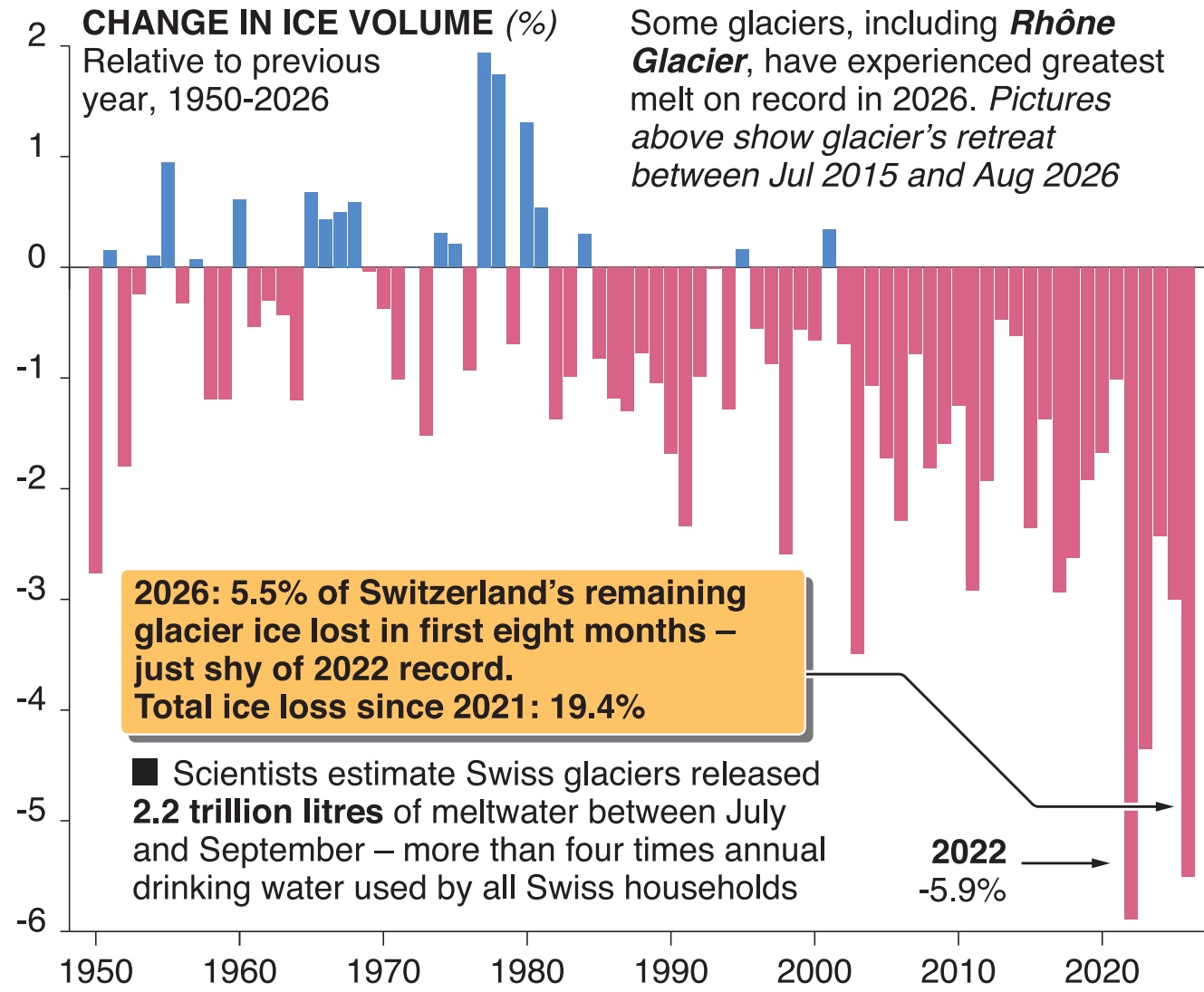
$x = 600 - 530$

$x = 70$

Swiss glaciers have endured a near-record loss of ice in 2026 following a summer of intense heatwaves, raising concerns about future water supplies for agriculture, hydropower and transport



Relative to previous
year, 1950-2026



Some glaciers, including **Rhône Glacier**, have experienced greatest melt on record in 2026. *Pictures above show glacier's retreat between Jul 2015 and Aug 2026*

2026: 5.5% of Switzerland's remaining glacier ice lost in first eight months – just shy of 2022 record.
Total ice loss since 2021: 19.4%

■ Scientists estimate Swiss glaciers released **2.2 trillion litres** of meltwater between July and September – more than four times annual drinking water used by all Swiss households

2022
-5.9%

Source: Swiss Glacier Monitoring Network (GLAMOS) Pictures: Getty Images © GRAPHIC NEWS

QWIKIPEDIA: A DAILY SEARCH-AND-LEARN ACTIVITY

Ever been on Wikipedia to search for something? More often than not we start with one topic, click a few links and within no time, are eagerly reading something else. Qwikipedia is a game that tries to quench our curiosity to learn something new. Start with one search from a specific article in this newspaper, and get started on a journey of learning with multiple clicks. Surprise yourself by finding new routes each time!

THE SATURDAY SAUNTER: HOT AIR BALLOON → PAMPAS



HOT AIR BALLOON

A hot air balloon is the oldest successful human-carrying flight technology, operating on the basic thermodynamic principle that warm air rises because it is less dense than cooler air. By heating the air inside a massive fabric envelope with a liquid propane burner, a pilot creates buoyancy (lift) that allows the aircraft to ascend into the sky. The first unanchored human voyage in a hot air balloon occurred in Paris, France, undertaken by Jean-François Pilâtre de Rozier alongside François Laurent d'Arlandes on November 21, 1783. There are three fundamental components of a hot air balloon: the envelope, the burner and the basket. The basket is typically made from solid stems of **rattan**.

RATTAN

Rattan is a naturally flexible, fibrous material derived from roughly 600 species of Old World climbing palms belonging to the subfamily Calamoideae. They are predominantly native to the tropical jungles of Southeast Asia (with Indonesia producing around 80% of the world's supply. Rattan is extensively used for making handicrafts like baskets, furniture, and decorative art. Generally, raw rattan is processed into several products to be used as materials. Rattan is also used to make mallets or sticks in **polo**.



POLO

Polo is a game played on horseback between two teams of four players. Each team uses mallets with long, flexible handles to drive a wooden ball down a grass field and between two goalposts. It is among the oldest equestrian sports. The origins of polo are uncertain. Some historians trace its roots to Central Asia, whereas others suggest that the horseback sport of sagol kangjei, played in the present-day Indian state of Manipur, dates back as far as 3100 BCE. Polo was brought to America and became hugely popular in Argentina and they were played by immigrants in the Argentine **pampas**.

PAMPAS

The Pampas Plains are fertile, low-lying grasslands covering over 12 lakh sq.km across southern South America. They span the Argentine provinces of Buenos Aires, La Pampa, Santa Fe, Entre Rios, and Cordoba, as well as the entirety of Uruguay and the southernmost state of Brazil, Rio Grande do Sul. Characterised by a temperate climate and deep, rich soil, the region serves as the agricultural heartland of Argentina, heavily utilised for crop farming and cattle grazing. Anthropogenic factors such as environmental changes, pollution and ecosystem disruptions caused by human activities have impacted the native fauna of the Pampas. A substantial number of mid-to-large-scale species, such as the puma, rhea, and capybara, have suffered severe habitat destruction driven by agricultural expansion and ranching practices, surviving today in only isolated pockets. Meanwhile, iconic creatures including the Jaguar and Guanaco have faced complete local extinction within this biome.



So, the path we have travelled today is Hot air balloons → Rattan → Polo → Pampas
Three clicks! Hop over!
Send in your paths for the above to school@thehindu.co.in with the subject: Qwikipedia

PHOTOS: REUTERS, WIKIMEDIA COMMONS

IN SCHOOL INBOX

Showcase your creativity

A couple of artivity entries sent in by students

ARTIVITY

Read the newspaper and create an original artwork based on a news item that interests you and send your entry in JPEG format (not as PDF) to school@thehindu.co.in with the subject - Artivity. ONLY artworks based on current news or issues relating to Science, Space, Environment, Wildlife and Sports will be considered for publication. Please AVOID mailing other works. Mention your name, class, school and city in the same mail (not in separate mails).



Name: K Pranav
Class :VII
DAV School, IIT Hyderabad Campus
Kandi, Sangareddy

SPLASH OF COLOURS



Members of the Panghat Performing Arts Group perform during a full dress rehearsal for the upcoming 'Navratri' festival, in Ahmedabad, Gujarat, on Oct. 4, 2026. PHOTO: PTI

GUESS WHO?



He is an American businessman and computer scientist who co-founded Google . He was the executive chairman of the company from 1997 until August 2001. From 2011 to 2015, he was appointed as the chairman of Google's newly formed parent company Alphabet Inc.

Answer: Larry Page

SUDOKU

5	8	3			4	2	
			4		1	7	5
			5				
			6		3	5	
7	9					8	4
4	5		2				
				6			
9	2	4		8			
	8	3			5	7	4

DIFFICULTY RATING: ★☆☆☆☆

A mind game and a puzzle that you solve with reasoning and logic. Fill in the grid with digits in such a manner that every row, every column and every 3x3 box accommodates the digits 1 to 9, without repeating any. The solution to today's puzzle is at right.

1	4	7	9	6	2	3	8	9
3	9	5	1	8	7	4	2	6
2	6	8	9	3	4	7	1	5
6	1	9	7	2	8	5	4	3
4	7	3	2	1	5	9	6	7
2	8	4	9	6	1	7	8	
8	3	5	7		2	9	1	
8	7	1	5	1	2	9	4	
5	4	6	8	1	7	5	2	
6	9	3	7	4	2	1	5	



India's Mandeep Singh, 11, Sanjay, 70, and others make an appeal during the men's final hockey match between India and Malaysia, at the Asian Games, in Kakamigahara, Japan, on, Oct. 3, 2026. PHOTO: PTI

After Asian Games feat, India hockey team turns focus to LA Olympics

The Olympic qualification, secured by winning the Asian Games, has given India valuable time to prepare for Los Angeles Olympics 2028

India's men's hockey team has secured a direct ticket to the 2028 Los Angeles Olympics and skipper Harmanpreet Singh and his teammates are looking ahead, insisting that the focus will remain on the process and using the next 18 months to build towards the Games.

Harmanpreet was delighted with the team's gold medal-winning campaign in Nagoya but said the players were repeatedly reminded to stick to their process and not get distracted by the scoreline.

"Very happy that we have returned with the medal, so no

need to say anything else," Harmanpreet said. "Stick to the process and training, and don't take any pressure, doesn't matter what the score is."

The Olympic qualification, secured by winning the Asian Games, has given India valuable time to prepare for Los Angeles, with Harmanpreet confident of making the most of the period leading up to the Games.

"We have a good opportunity that we will get good time before LA Olympics and looking forward that we will do good there and in all the matches before that as well," he said.

India had to dig deep against Pakistan in a tense semifinal, with Harmanpreet admitting that the team made some mistakes but managed to hold its nerve and secure the win.

"The match went really well, it was one such match of the tournament where we did some silly mistakes and I think somehow that last minute goal was very good but in defense overall the whole team performed well," he said.

Veteran mid-fielder Manpreet Singh acknowledged that Pakistan's attacking quality made the semifinal a difficult

contest, but said India always believed they could prevail if they played at their best.

"It was a good match, because they are a good team, they have got good strikers, they punished us on our mistakes as well, but we knew that we will definitely win if we will give our 100 per cent," he said. For forward Mandeep Singh, the Asian Games had been the primary target after India's World Cup campaign.

"The performance in the World Cup, it's not that we didn't get the medal but the aim was that we had to qualify for the LA Olympics in the Asian Games," he said. rti

'Paris Olympics bronze was the most satisfying moment'

India defended its gold medal in style at the just-concluded Asian Games but men's team head coach Craig Fulton still rates the bronze medal finish at the Paris Olympics as the most satisfying achievement during his current tenure.

Reigning Olympic bronze medallist India hammered Malaysia 5-1 in the final and secured a direct qualification for the 2028 Los Angeles Olympics.

"Most satisfying moment is probably the Paris Olympics in a way of how we had stuck to our structures and our processes, even though we were losing games in the FIH Pro League and losing games against Australia before the Paris Olympics," Fulton told PTI in an exclusive interview.



Craig Fulton. FILE PHOTO THE HINDU

"And then we peaked at the right time through good planning and obviously with all the staff doing a great job. And what happened then is we beat Australia for the first time in 50 years."

The Indian men's hockey team defeated Australia 3-2 at the Paris Olympics, ending a 52-year Olympic win drought against the Kookaburras.

"Then we go 10 men down and we managed to get a shootout win. We draw the game with Great Britain and get a shootout win in the quarters. And then we play a really good game against Germany in the semifinals and just lose 3-2."

Fulton said it was a very difficult period but he is delighted with the fact that India have dominated Asian hockey.

"Yeah, that period was very special. But at the same time, in Asia, we've had an amazing run. People say it's easy to win back-to-back medals in the Asia Games. It's never been done before. India has never done it. So, it's not easy. I'm very proud of the performance that we've done in Asia," he said.

With only 20 days between FIH World Cup, where India finished a disappointing eighth, and the Asian Games, it was not a cakewalk for Fulton and his men.

'Sticking with World Cup squad'

Fulton said he decided to stick to his core team for both the World Cup and the Asian Games was a conscious decision, even though there were hardly three weeks between the two events.

"Well, you see, if you have a different team or a different tournament, who's going to coach them? Who's going to prepare them? While I am at the World Cup, who's coaching this group of players with the same tactics, with the same mindset and what we're trying to do? That's why it was easier to take one squad of players, 22 squad of players, bring them to Holland, train everyone there while we were playing in the World Cup. The other boys were still training," Fulton said. Fulton emphasised upon the need for transition in Indian men's team looking forward to LA 2028 and said it will start from the Asian Champions Trophy (ACT) to be held in Mohali and Jalandhar from October 27 to November 5.

"We'll have 30 players there. We'll choose a squad of 20 to play the ACT and then extra players will stay on because they'll be used to go and play against New Zealand in New Zealand a week after the ACT. So, again, we'll use another group of players," he said. rti

HAVE YOU READ TODAY'S IN SCHOOL ?

Here's how you can engage yourself in the learning process with the newspaper. Maintain a book and write down the activities date-wise. Besides honing your language skills, it will also help you sharpen your critical thinking and relate better to the world around you.

THE THREE WORDS I LEARNT TODAY



Words	Meanings
1).....	
2).....	
3).....	

BACKGROUND CHECK

Select an article that interests you, do some research on the subject and write down three interesting points on the subject.

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PICK A PICTURE, TELL THE STORY....

Go through the newspaper, select one picture, and write five lines about it.

Picture / Page number



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LEARNING CURVE

Two facts that I learned from today's newspaper.

1.....

2.....



MY DAILY 8 TO 8 PLANNER

Use this chart to plan your day.



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MY MOOD BOARD

Draw an emoji to express how you feel today. Get as creative as possible.



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MY GRAMMAR PINS

Read the newspaper and answer the following.

NOUNS & VERBS
Pick five words and tick them as nouns or verbs.

.....

noun/ verb

.....

noun/ verb

.....

noun/ verb

.....

noun/ verb

.....

noun/ verb

PAST, PRESENT, FUTURE

Pick two words each in the Past, Present and Future tenses.

PAST

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PRESENT

.....

FUTURE

.....